

Online Supplementary Materials

S.1 Proof of Proposition 6

With the generalized form of consumers' psychological responses to price differences, the platform's total profit is

$$\Pi^G = p_H D_H^G(p_H, p_L) N(f) + p_L D_L^G(p_H, p_L) N(f).$$

The first-order conditions for profit maximization with respect to p_H and p_L are

$$\frac{\partial \Pi^G}{\partial p_H} = 0 \Leftrightarrow D_H^G + (p_H + f) \frac{\partial D_H^G}{\partial p_H} + (p_L + f) \frac{\partial D_L^G}{\partial p_H} = 0, \quad (\text{S1})$$

$$\frac{\partial \Pi^G}{\partial p_L} = 0 \Leftrightarrow (p_H + f) \frac{\partial D_H^G}{\partial p_L} + D_L^G + (p_L + f) \frac{\partial D_L^G}{\partial p_L} = 0. \quad (\text{S2})$$

Solving the simultaneous equations gives the best responses of the consumer-side prices to the price for sellers in equilibrium:

$$p_H^*(f^*) = \frac{(2 + 2\eta)\beta_H + (\eta + \gamma)(1 + \beta_L)}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} - \frac{2 + 3\gamma + \eta + 2\gamma\eta - \gamma^2 - \eta^2}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} f^*, \quad (\text{S3})$$

$$p_L^*(f^*) = \frac{(\eta + \gamma)\beta_H + (2 + 2\gamma)(1 + \beta_L)}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} - \frac{2 + \gamma + 3\eta + 2\gamma\eta - \gamma^2 - \eta^2}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} f^*. \quad (\text{S4})$$

Thus, the equilibrium prices such that $p_H > p_L$ satisfy

$$p_H^*(f^*) - p_L^*(f^*)(f) = B^G + T^G f^*,$$

with $B^G = \frac{(2+\eta-\gamma)(1+\beta_H)-(2-\eta+\gamma)(1+\beta_L)}{4+4\gamma+4\eta-2\gamma\eta-\gamma^2-\eta^2}$, and $T^G = -\frac{2\gamma-2\eta}{4+4\gamma+4\eta-2\gamma\eta-\gamma^2-\eta^2}$.

Here, given $\gamma > \eta > 0$, we have $B^G > B > 0$ and

$$\frac{\partial T^G}{\partial \eta} = \frac{8 + 16\eta - 4\gamma\eta + 2\gamma^2 - 6\eta^2}{(4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2)^2} > 0.$$

Consider that $T^G = T$ at $\eta = 0$. We further have $T < T^G < 0$.

S.2 Proof of Proposition 7

When the proportions of high-type and low-type consumers can be different, the platform's total profit becomes

$$\tilde{\Pi} = \frac{2\mu}{1+\mu}(p_H + f)\widetilde{D_H}(p_H, p_L)N(f) + \frac{2}{1+\mu}(p_L + f)D_L(p_L)N(f).$$

For maximizing the platform's profit, the first-order condition with respect to p_H has the same form with equation (6), while the first-order condition with respect to p_L becomes

$$\frac{\partial \tilde{\Pi}}{\partial p_L} = 0 \Leftrightarrow \mu(p_H + f)\frac{\partial D_H}{\partial p_L} + D_L + (p_L + f)\frac{\partial D_L}{\partial p_L} = 0. \quad (\text{S5})$$

Solving the simultaneous equations gives the best responses of the consumer-side prices to the price for sellers in equilibrium:

$$p_H^*(f^*) = \frac{2(1 + \beta_H) + \gamma(1 + \beta_L)}{4 + 4\gamma - \mu\gamma^2} - \frac{2 + 3\gamma - \mu\gamma^2}{4 + 4\gamma - \mu\gamma^2}f^*, \quad (\text{S6})$$

$$p_L^*(f^*) = \frac{\mu\gamma(1 + \beta_H) - 2(1 + \beta_L)}{4 + 4\gamma - \mu\gamma^2} - \frac{2 + (2 - \mu)\gamma - \mu\gamma^2}{4 + 4\gamma - \mu\gamma^2}f^*. \quad (\text{S7})$$

Thus, the equilibrium prices such that $p_H > p_L$ satisfy

$$p_H^*(f^*) - p_L^*(f^*) = \tilde{B} + \tilde{T}f^*,$$

with $\tilde{B} = \frac{(2-\mu\gamma)(1+\beta_H)-(2+\gamma)(1+\beta_L)}{4+4\gamma-\mu\gamma^2}$ and $\tilde{T} = -\frac{(1+\mu)\gamma}{4+4\gamma-\mu\gamma^2}$. Here, it is easy to find that $\tilde{B}$ is positive and $\tilde{T}$ is negative as $p_H > p_L$ on the interior, and we also have

$$\frac{\partial \tilde{T}}{\partial \mu} = -\frac{4\gamma + 4\gamma^2 + \gamma^3}{(4 + 4\gamma - \mu\gamma^2)^2} < 0.$$

S.3 Discussion on endogenous γ to μ

Here, we further extend the demand of high-type consumers to

$$\widetilde{D}_H(p_H, p_L) = 1 + \beta_H - p_H - \gamma(\mu)(p_H - p_L),$$

where $\mu = \frac{\lambda}{1-\lambda} \in (0, +\infty)$, $\gamma(0) = \gamma_0 \in (0, 1)$, and $\lim_{\mu \rightarrow +\infty} \gamma(\mu) = 0$. We assume $\frac{\partial \gamma(\mu)}{\partial \mu} = 0$ for $\mu < 1$ and $\frac{\partial \gamma(\mu)}{\partial \mu} = \alpha < 0$ for $\mu \geq 1$. This formulates how consumers' aversion to pricing unfairness is endogenous to the proportion of high-type consumers. Recall

$$p_H^*(f^*) - p_L^*(f^*) = \widetilde{B} + \widetilde{T}f^*,$$

with $\widetilde{B} = \frac{(2-\mu\gamma)(1+\beta_H)-(2+\gamma)(1+\beta_L)}{4+4\gamma-\mu\gamma^2}$ and $\widetilde{T} = -\frac{(1+\mu)\gamma}{4+4\gamma-\mu\gamma^2}$. In the case of $\gamma(\mu)$, we derive

$$\frac{\partial \widetilde{T}}{\partial \mu} < 0 \Leftrightarrow \frac{\partial \gamma(\mu)}{\partial \mu} > -\frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)},$$

$$\frac{\partial \widetilde{T}}{\partial \mu} \geq 0 \Leftrightarrow \frac{\partial \gamma(\mu)}{\partial \mu} \leq -\frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)}.$$

For $\mu < 1$, given that $\frac{\partial \gamma}{\partial \mu} = 0$, there must be

$$\frac{\partial \gamma(\mu)}{\partial \mu} = 0 > -\frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)}.$$

For $\mu \geq 1$, given that $\frac{\partial \gamma}{\partial \mu} = \alpha < 0$, there must be

$$\frac{\partial \gamma(\mu)}{\partial \mu} = \alpha < \lim_{\mu \rightarrow +\infty} \frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)} = 0.$$

In this case, it is uncertain how the trade-off between the price difference on the consumer side and the price for sellers changes as μ increases. On the one hand, adding weight to high-type consumers still makes it more profitable to attract high-type consumers relative to low-type consumers. This strengthens the demand externality from low-type to high-type consumers. On the other hand, in the case where high-type consumers are in the majority (i.e., $\mu \geq 1$), a higher μ additionally implies that consumers' aversion to pricing unfairness gets weaker (i.e., a lower γ). This weakens the demand externality and thus weakens the trade-off.

Here we mark two special cases. First, when high-type consumers are in the minority (i.e., $\mu < 1$), the positive effect of increasing the proportion on the trade-off is definitely greater than the negative effect. Second, when the proportion of high-type consumers is extremely large (i.e., $\mu \rightarrow +\infty$), the positive effect of increasing the proportion on the trade-off is smaller than the negative effect.

S.4 Proofs of Lemma 3 and Proposition 8

When the platform can implement seller-side price discrimination, its total profit becomes

$$\begin{aligned}\widehat{\pi}(p_H, p_L, f_H, f_L) = & \frac{1}{2}[(p_H + f_H)D_H(p_H, p_L) + (p_L + f_H)D_L(p_L)]N_H(f_H) + \\ & \frac{1}{2}[(p_H + f_L)D_H(p_H, p_L) + (p_L + f_L)D_L(p_L)]N_L(f_L).\end{aligned}$$

The FOCs are

$$\begin{aligned}\frac{\partial \Pi}{\partial p_H} = 0 & \Leftrightarrow p_H = \frac{1 + \beta_H}{2(1 + \gamma)} + \frac{\gamma}{2(1 + \gamma)}p_L - \frac{1}{2} \frac{f_H N_H + f_L N_L}{(N_H + N_L)}, \\ \frac{\partial \Pi}{\partial p_L} = 0 & \Leftrightarrow p_L = \frac{1 + \beta_L}{2} + \frac{\gamma}{2}p_H - \frac{1 + \gamma}{2} \frac{f_H N_H + f_L N_L}{(N_H + N_L)}, \\ \frac{\partial \Pi}{\partial f_H} = 0 & \Leftrightarrow f_H = \frac{1 + \theta_H}{2} - \frac{1}{2} \frac{p_H D_H + p_L D_L}{(D_H + D_L)}, \\ \frac{\partial \Pi}{\partial f_L} = 0 & \Leftrightarrow f_L = \frac{1 + \theta_L}{2} - \frac{1}{2} \frac{p_H D_H + p_L D_L}{(D_H + D_L)}.\end{aligned}$$

Define the consumer-side and seller-side average prices as

$$\bar{p} = \frac{p_H D_H + p_L D_L}{(D_H + D_L)} \quad \text{and} \quad \bar{f} = \frac{f_H N_H + f_L N_L}{(N_H + N_L)}.$$

Plugging the average prices into the above FOCs gives the following results on the interior:

$$f_L^*(\bar{p}^*) - f_H^*(\bar{p}^*) = \frac{\theta_L - \theta_H}{2},$$

$$p_H^*(\bar{f}^*) - p_L^*(\bar{f}^*) = \widehat{B} + \widehat{T}\bar{f}^*,$$

with $\widehat{B} = B = \frac{(2-\gamma)(1+\beta_H)-(2+\gamma)(1+\beta_L)}{4+4\gamma-\gamma^2} > 0$ and $\widehat{T} = T = -\frac{2\gamma}{4+4\gamma-\gamma^2} < 0$.

S.5 Welfare Issues

Mandatory uniform pricing. Supposing that uniform pricing is on the interior, high-type consumers' surplus, low-type consumers' surplus, sellers' surplus, and total surplus are given as

$$CS_H^- = \int_p^{1+\beta_H} \int_f^{1+\theta} (v_H - p) dv_S dv_H = \frac{1}{2}(1 + \theta - f)(1 + \beta_H - p)^2,$$

$$CS_L^- = \int_p^{1+\beta_L} \int_f^{1+\theta} (v_L - p) dv_S dv_L = \frac{1}{2}(1 + \theta - f)(1 + \beta_L - p)^2,$$

$$\begin{aligned} RS^- &= \int_f^{1+\theta} \int_p^{1+\beta_H} (v_S - f) dv_H dv_S + \int_f^{1+\theta} \int_p^{1+\beta_L} (v_S - f) dv_L dv_S \\ &= \frac{1}{2}[(1 + \beta_H - p) + (1 + \beta_L - p)](1 + \theta - f)^2, \end{aligned}$$

$$\begin{aligned} TW^- &= CS_H^- + CS_L^- + RS^- + \Pi^- \\ &= N \left\{ \frac{(1 + \beta_H)^2 - p^2}{2} + \frac{(1 + \beta_L)^2 - p^2}{2} + \frac{(D_H + D_L)(1 + \theta + f)}{2} \right\}. \end{aligned}$$

Price discrimination without consumers' fairness concerns. On the interior, high-type consumers' surplus, low-type consumers' surplus, sellers' surplus, and total surplus are given as

$$CS_H^\circ = \int_{p_H}^{1+\beta_H} \int_f^{1+\theta} (v_H - p_H) dv_S dv_H = \frac{1}{2}(1 + \theta - f)(1 + \beta_H - p_H)^2,$$

$$CS_L^\circ = \int_{p_L}^{1+\beta_L} \int_f^{1+\theta} (v_L - p_L) dv_S dv_L = \frac{1}{2}(1 + \theta - f)(1 + \beta_L - p_L)^2,$$

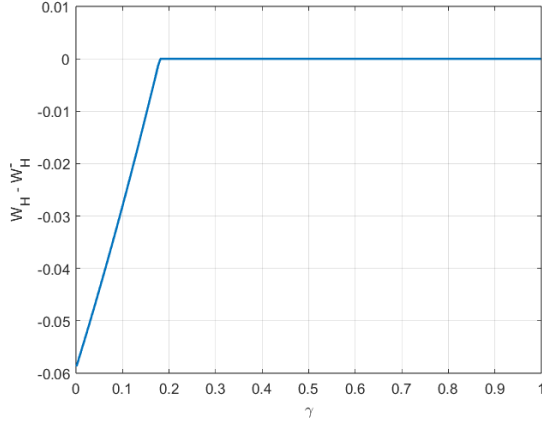
$$\begin{aligned} RS^\circ &= \int_f^{1+\theta} \int_{p_H}^{1+\beta_H} (v_S - f) dv_H dv_S + \int_f^{1+\theta} \int_{p_L}^{1+\beta_L} (v_S - f) dv_L dv_S \\ &= \frac{1}{2}[(1 + \beta_H - p_H) + (1 + \beta_L - p_L)](1 + \theta - f)^2, \end{aligned}$$

$$\begin{aligned} TW^\circ &= CS_H^\circ + CS_L^\circ + RS^\circ + \Pi^\circ \\ &= N \left\{ \frac{(1 + \beta_H)^2 - p_H^2}{2} + \frac{(1 + \beta_L)^2 - p_L^2}{2} + \frac{(D_H + D_L)(1 + \theta + f)}{2} \right\} \end{aligned}$$

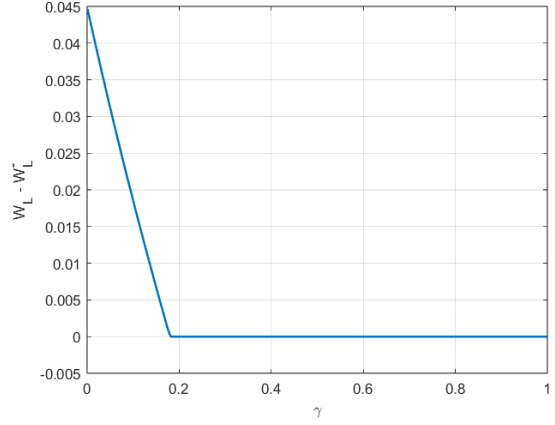
Price discrimination with consumers' fairness concerns. On the interior, high-type consumers' surplus, low-type consumers' surplus, sellers' surplus, and total surplus are given as

$$\begin{aligned}
CS_H &= \int_{p_H + \gamma(p_H - p_L)}^{1 + \beta_H} \int_f^{1 + \theta} [v_H - p_H - \gamma(p_H - p_L)] dv_S dv_H \\
&= \frac{1}{2} (1 + \theta - f) [1 + \beta_H - p_H - \gamma(p_H - p_L)]^2, \\
CS_L &= \int_{p_L}^{1 + \beta_L} \int_f^{1 + \theta} (v_L - p_L) dv_S dv_L = \frac{1}{2} (1 + \theta - f) (1 + \beta_L - p_L)^2, \\
RS &= \int_f^{1 + \theta} \int_{p_H + \gamma(p_H - p_L)}^{1 + \beta_H} (v_S - f) dv_H dv_S + \int_f^{1 + \theta} \int_{p_L}^{1 + \beta_L} (v_S - f) dv_L dv_S \\
&= \frac{1}{2} \{ [1 + \beta_H - p_H - \gamma(p_H - p_L)] + (1 + \beta_L - p_L) \} (1 + \theta - f)^2, \\
TW &= CS_H + CS_L + RS + \Pi \\
&= N \left\{ \frac{(1 + \beta_H)^2 - [p_H + \gamma(p_H - p_L)]^2}{2} + \frac{(1 + \beta_L)^2 - p_L^2}{2} + \frac{(D_H + D_L)(1 + \theta + f)}{2} - \right. \\
&\quad \left. \gamma(p_H - p_L) D_H \right\}.
\end{aligned}$$

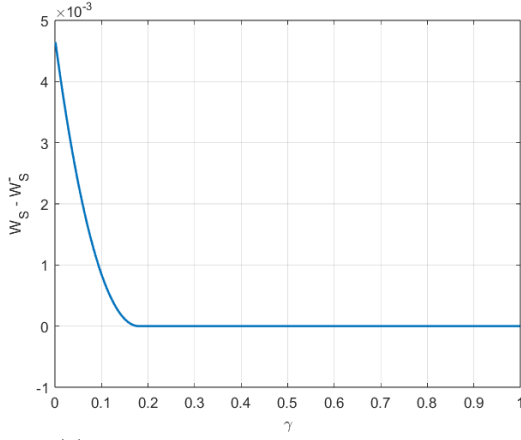
According to the total surplus expressions, a positive fairness concerns term (i.e., $\gamma(p_H - p_L) > 0$) always induces a type of “deadweight loss” to the social planner. Hence, the social optimum on the interior can be achieved only if the fairness concerns term is equal to 0. Comparing TW^- and TW° with TW , it is easy to find that $TW^- = TW^\circ > TW$ holds for price discrimination on the interior. For illustration, we present welfare simulations under price-discrimination with consumers’ fairness concerns in comparison with the other two cases. The results are reported in Supplementary Figures [S1](#) and [S2](#).



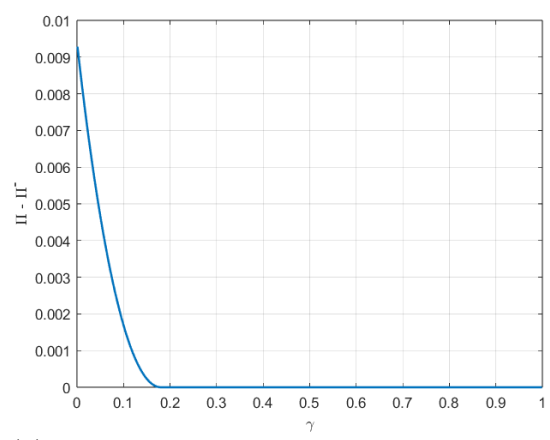
(a) Relative change in high-type consumers' surplus



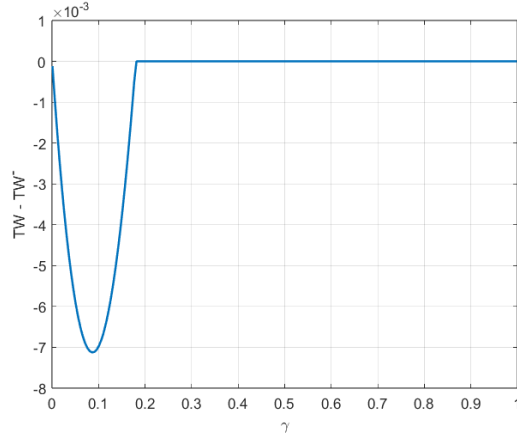
(b) Relative change in low-type consumers' surplus



(c) Relative change in sellers' surplus

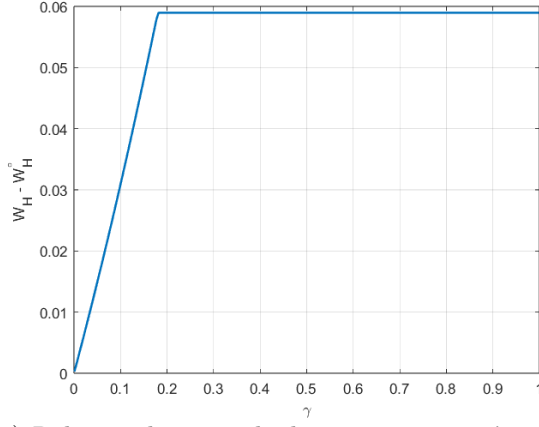


(d) Relative change in the platform's total profit

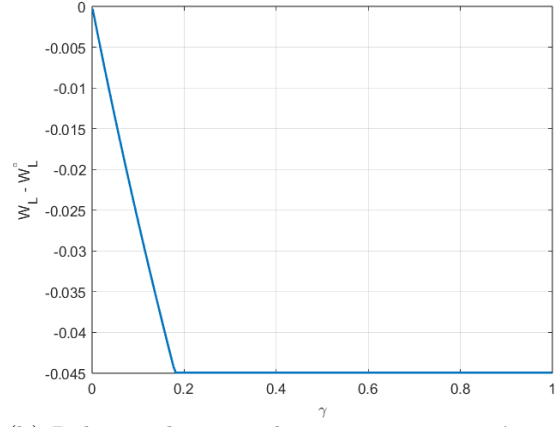


(e) Relative change in total surplus

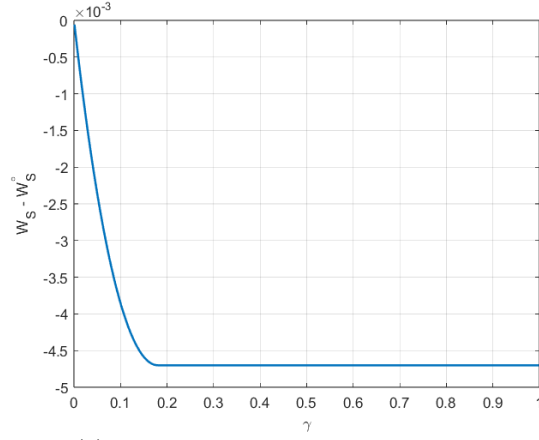
Figure S1: Welfare comparison between the cases of price discrimination with consumers' fairness concerns and of mandatory uniform pricing. Parameter values: $\beta_H = 0.4$, $\beta_L = 0.1$, $\theta = 0.25$, and $\gamma \in (0, 1)$.



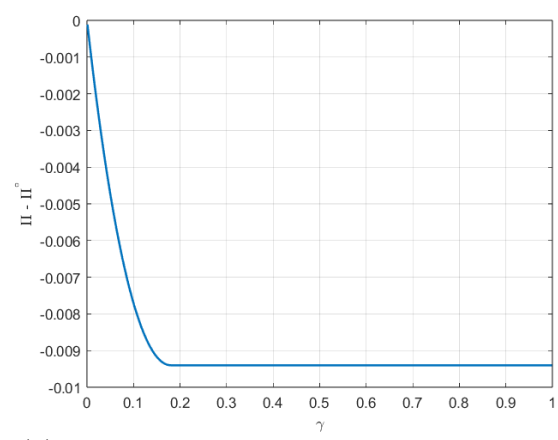
(a) Relative change in high-type consumers' surplus



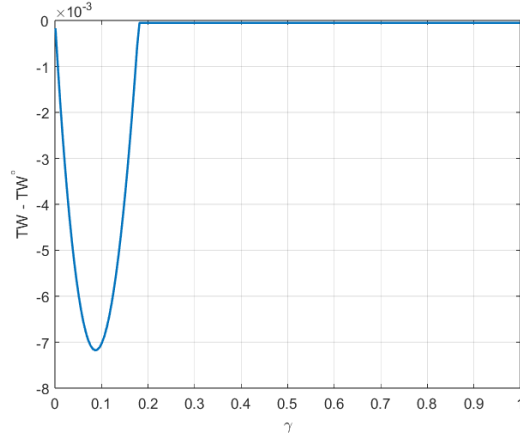
(b) Relative change in low-type consumers' surplus



(c) Relative change in sellers' surplus



(d) Relative change in the platform's total profit



(e) Relative change in total surplus

Figure S2: Welfare comparison between the cases with and without consumers' fairness concerns. Parameter values: $\beta_H = 0.4$, $\beta_L = 0.1$, $\theta = 0.25$, and $\gamma \in (0, 1)$.